

Special Session Proposal - E2A 2027

Title of the Proposal: ASCVA-UAV: Advances in Sensing, Control and Vision for Secure Autonomy in UAVs

A Brief Description of the Proposed Area

Unmanned Aerial Vehicles (UAVs) have rapidly transitioned from niche defence platforms to mainstream tools for precision agriculture, disaster response, infrastructure inspection, surveillance, logistics and environmental monitoring. India's UAV ecosystem is expanding under national policy thrusts such as the Production-Linked Incentive (PLI) scheme for UAVs and the drive toward Atmanirbhar Bharat, creating a strong need for indigenous, low-cost, and reliable UAV subsystems that reduce dependence on imported sensors, controllers and vision modules.

A UAV platform is fundamentally a tightly coupled chain of subsystems: (i) the physical sensors that perceive the vehicle's state and environment, (ii) the analog front-end that conditions and digitizes these raw sensor signals with adequate accuracy and noise immunity, (iii) the flight controller that uses this sensed information to compute stabilizing and trajectory-tracking commands in the presence of strong nonlinearities and disturbances, (iv) the onboard or ground-based vision system that extracts actionable information from captured images and video for navigation, mapping and situational awareness, and (v) the security layer that protects sensor data, the command-and-control link, and onboard autonomy from spoofing, jamming, GPS/GNSS deception and cyber-intrusion. Most existing academic treatments address these layers in isolation. This session is deliberately structured to bring them together as a single, indigenously realizable and secure design chain, from silicon/PCB-level sensor fabrication up to real-time image processing and system-level security, with cost, size, weight and power (C-SWaP) as an explicit design constraint throughout.

The session welcomes theoretical, simulation-based, and hardware-validated contributions, and is expected to attract researchers from instrumentation, control systems, VLSI/MEMS sensor fabrication, embedded systems, robotics and computer vision communities, fostering interdisciplinary collaboration around a application area of high current relevance to both academia and industry.

Rationale for Organizing this Special Session

UAV research is typically presented in a fragmented manner across separate instrumentation, control, and vision-related tracks, making it difficult for authors and attendees working on a specific UAV subsystem to see how their contribution fits into a deployable, indigenous platform. This session is proposed to give the growing community working on low-cost UAV technology in India a single, focused venue that spans the complete design chain -- sensor fabrication, signal conditioning, nonlinear control, image processing, and security -- rather than treating these as isolated topics under generic tracks.

The timing is also significant: national initiatives such as the Production-Linked Incentive (PLI) scheme for drones and the push toward indigenous, Atmanirbhar UAV manufacturing are creating strong demand for exactly this kind of cross-layer, cost-aware research. The organizing team's own complementary expertise across sensing, instrumentation, control and vision reflects the interdisciplinary nature of the session and positions it to attract quality submissions and reviewers from each of these sub-communities, while giving delegates a coherent, systems-level perspective on UAV design that the regular tracks do not offer.

Details of Session Organizers:

Organizer 1

Name: Dr. Deep Chandra Upadhyay

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Photo:



Brief bio: Dr. Deep Chandra Upadhyay is currently working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I completed my Ph.D. in the Department of Electronics Engineering at the Indian Institute of Technology (BHU), Varanasi, with a specialization in Nanoelectronics. I earned my M.Tech in Nano Science and Technology from the Indian Institute of Technology Patna and my B.Tech. in Electronics and Communication Engineering.

I also worked as a Research Assistant under the guidance of Prof. B. V. R. Chowdari as part of the SIPIS Program (2013) at the National University of Singapore (NUS). I have over five years of academic experience and have been actively engaged in research projects focused on nanoscale device fabrication, optoelectronic sensors, and digital integrated circuit design. I have authored and co-authored more than 30 research publications in reputed international journals, including Q1 and IEEE journals, as well as international conferences. I have also actively contributed to the organization of international conferences, serving as the General Co-Chair of the 2026 IEEE International Conference on Intelligent Electronics and Computational Systems (IECS 2026) and as the Co-Convener of the International Conference on Signal and Data Processing (ICSDP-2024).

Organizer 2

Name: Dr. Anurag Tiwari

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Brief bio: Dr. Anurag Tiwari is currently working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I received my B.Tech. degree in Electronics and Communication Engineering from Rewa Engineering College, Rewa, India, and my M.Tech degree in Digital Systems from Motilal Nehru National Institute of Technology (MNNIT),

Allahabad, India. I completed my Ph.D. at MNNIT Allahabad, India, with a specialization in Digital Image Watermarking.

I am actively engaged in research focusing on digital image watermarking, data security, and multimedia security. My research interests include Signal and Image Processing, Digital Image Watermarking, Data Hiding, Cryptography, Machine Learning, and Deep Learning. I have authored and co-authored more than 20 research publications in reputed international journals, conferences, and book chapters. I also actively contribute to the organization of international conferences and serve as the Co-Convener of the International Conference on Signal and Data Processing (ICS DP-2025).

Organizer 3

Name: Dr. Gopal Singh

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Brief bio: Dr. Gopal Singh works is working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University. I completed my Ph.D. in the Department of Instrumentation and Control Engineering at the National Institute of Technology, Tiruchirappalli, with a specialization in Electronic Instrumentation. I earned my M.Tech in Control and Instrumentation from the Delhi Technological University, New Delhi and my B.Tech. in Instrumentation and Control Engineering.

I am actively engaged in research focusing on sensor interfacing, electronic instrumentation, and precision measurement systems. My research interests include Sensor Interface Circuits, Electronic Instrumentation and Measurement, Analog and Mixed-Signal Circuit Design, Resistive and Thermoelectric Sensors. I have authored and co-authored several research publications in reputed international journals, including IEEE Transactions journals, as well as international conferences. My work has also resulted in a granted Indian patent in the field of sensor instrumentation. I am currently serving as an Associate Editor of the Journal of Electrical and Electronics Engineering (JEEE).

Organizer 4

Name: Dr. Dipak Prasad

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Brief bio: Dr. Dipak Prasad is currently working as an Assistant Professor in the School of Electrical and Electronics Engineering (SEEE) at VIT Bhopal University, India. I received my B.Tech. degree in Electrical Engineering from Maulana Abul Kalam Azad University of Technology (formerly WBUT), West Bengal, India, in 2015, and my M.Tech degree in Control and Industrial Automation from the National Institute of Technology (NIT) Silchar, Assam, India, in 2019. I completed my Ph.D. in Control Systems from Delhi Technological University (DTU), Delhi, India, in 2025.

I have published research articles in reputed Q1 and Q2 peer-reviewed journals and IEEE conferences. I am actively engaged in research focusing on nonlinear control systems, singularly perturbed and switched systems, intelligent control, robotics, and AI-enabled automation. My research interests include Nonlinear Dynamical Systems, Adaptive and Robust Control, Neural Network-Based Control, Disturbance Observer Design, Robotics, Electric Vehicle Control Systems, Digital Twin Technologies, and AI-Driven Autonomous and Intelligent Control Systems. I have authored and co-authored several research publications in reputed international journals and conferences. Alongside my research endeavors, I actively mentor students in research, innovation, and industry-oriented project development, fostering analytical thinking, practical problem-solving skills, and interdisciplinary learning.

Technical Outline of the Session and Topics

The session is organized around the complete indigenous UAV design chain -- from sensor fabrication to onboard intelligence -- and invites original contributions in, but not limited to, the following areas:

- Low-cost, indigenous sensor fabrication and design for UAVs (inertial, altitude, proximity, strain/displacement and MEMS/piezoelectric sensing elements), including calibration and reliability under airborne operating conditions
- Sensor signal conditioning and instrumentation circuits (analog front-ends, ADC architectures, noise/EMI and error compensation) optimized for low power, low cost and weight-constrained UAV platforms
- Nonlinear, adaptive and fault-tolerant controller design for UAV attitude, altitude and trajectory control under disturbances, payload variation and actuator faults
- Sensor fusion and state estimation techniques (Kalman/extended Kalman/particle filtering) for UAV navigation and control
- Low-cost embedded/edge-based image and video processing for UAVs, including object detection, tracking, obstacle avoidance and visual navigation
- Security of UAV sensing, control and communication, including GNSS spoofing resilience, secure C2 links, and lightweight cryptography/intrusion detection for resource-constrained platforms
- System-level integration, real-time embedded architectures, and field validation of complete indigenous low-cost UAV platforms

Potential Reviewers

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